

Sensitivity analysis of normal mode algorithms using automatic differentiation

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ABSTRACT:

Acoustic propagation underwater is affected by dynamic and spatial variations of the sound speed, both of which can be observed in underwater environments. In this work, a differentiable implementation of the KRAKEN normal mode model is presented. By applying automatic differentiation to the discretized algebraic eigenvalue problem, we obtain sensitivities of every modal quantity with respect to a full set of geoacoustic parameters and frequency. These sensitivities are accurate to machine precision. This opens gradient-based methods for inversion purposes, such as source localization and geoacoustic inversion. The utility of this approach is demonstrated by performing sensitivity analyses on classical benchmark problems. These analyses provide insight into parameter significance, validate our implementation, and offer a foundation for developing more efficient gradient-based inversion techniques. © 2026 Acoustical Society of America. <https://doi.org/10.1121/10.0043870>

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I. INTRODUCTION

Acoustic propagation underwater is affected by temporal and spatial variations of the sound speed, both of which can be observed in underwater environments. Knowledge of the variation in sound speed is important for adequately modeling acoustic propagation in environments of interest (Jensen *et al.*, 2011). Accurate modeling of acoustic propagation is necessary for the solution of various inverse problems; for example, acoustic source localization, underwater communication, environmental monitoring, and evaluation (Duda, 2019). In shallow underwater environments, acoustic signals travel through both the water column and the seabed. Due to the seabed interaction, other geoacoustic properties become important, such as the density profiles of the seabed or compressional wave attenuation, in addition to the sound speed profile (SSP).

In both forward and inverse underwater acoustic problems, understanding how changes in environmental parameters affect acoustic propagation is crucial. This knowledge, often quantified through sensitivity analysis, informs the experimental design, identifies key parameters for inversion, and helps assess model uncertainty (Dosso *et al.*, 2007a; Dosso *et al.*, 2007b). Conceptually, the sensitivity of physical systems is most easily estimated using finite-difference (FD) methods, where the model is run repeatedly with perturbed parameters. However, this approach can be computationally prohibitive, especially for complex models or a large number of parameters, and may suffer from numerical inaccuracies related to truncation and round-off errors (Baydin *et al.*, 2015).

Quantifying uncertainty is also paramount, where estimated parameters inherently have associated uncertainties due to measurement noise and model limitations. Bayesian inference methods, often employing computationally intensive sampling techniques, such as Markov Chain Monte Carlo or specialized Gibbs samplers (Dosso, 2002), have become a standard approach for obtaining posterior probability distributions of parameters, providing a rigorous uncertainty quantification (Dettmer *et al.*, 2010; Dosso *et al.*, 2014). Theoretical analyses also explore the fundamental asymptotic accuracy limits achievable in geoacoustic inversions (Zanolin *et al.*, 2004). However, propagating these uncertainties through complex acoustic models remains a challenge, further highlighting the need for efficient sensitivity computation.

Beyond FDs, there are other methods for obtaining sensitivity information. The analytical calculation of derivatives offers greater accuracy and efficiency when feasible. For specific models such as normal modes, analytical or semi-analytical sensitivities have been derived based on variational or perturbation principles (Aki and Richards, 2002; Jensen *et al.*, 2011; Thode, 2004; Thode and Kim, 2004), including recent analytical formulations for the Pekeris-Airy waveguide (Petrov *et al.*, 2025) and dispersion equations for a shallow-water waveguide (Zakharenko *et al.*, 2026). In the broader field of geophysics, the analytical sensitivity kernel approach is widely utilized to investigate the sensitivity of modal dispersion curves to geoacoustic parameters (Pan *et al.*, 2019; Shi *et al.*, 2025). While these analytical kernels provide deep physical intuition by differentiating the continuous equations, they require complex manual derivations that must be redone if the underlying physics or boundary conditions change. In contrast, the automatic differentiation

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(AD) approach presented here differentiates the discrete numerical algorithm itself, offering the exact numerical gradient with extreme flexibility, as it automatically updates whenever the forward model is modified. For optimization-based inversion or data assimilation problems common in oceanography and Earth sciences, adjoint modeling methods provide an elegant and efficient way to compute the gradient of a misfit function with respect to many parameters by propagating the misfit backward through a linearized version of the forward model (Hermund *et al.*, 2006; Hursky *et al.*, 2004; Papadakis and Karasmani, 2020; Wunsch, 2006). More recently, techniques based on information geometry have also been explored to quantify the sensitivity and uncertainty of acoustic models (Spendlove *et al.*, 2024a; Spendlove *et al.*, 2024b).

With the recent rise of deep learning as an important framework (Goodfellow *et al.*, 2016; LeCun *et al.*, 2015), the need to compute gradients efficiently has become increasingly important, and AD is the core technology that allows training of complex neural networks (Baydin *et al.*, 2015). In this article, AD is used in a different context: Apply it directly to a physics-based propagation model by developing a differentiable implementation of an acoustic normal mode solver in the Julia programming language (Bezanson *et al.*, 2017). Currently, widely used normal mode solvers like KRAKEN and ORCA (Westwood *et al.*, 1996) are written in FORTRAN and lack built-in gradient capabilities, limiting their integration into modern gradient-based optimization and machine learning framework workflows.

The decision to enable AD by writing a differentiable normal mode solver is based on the distinct advantages AD presents over alternative methods, such as FD and symbolic differentiation, particularly for complex scientific computations (Baydin *et al.*, 2015). Unlike FD, which approximate derivatives and suffer from truncation errors, numerical instability, and a computational cost that scales linearly with the number of input parameters (requiring $N + 1$ function evaluations for N parameters), AD computes derivatives to machine precision. Crucially, AD allows the gradient vector to be calculated with a computational cost typically only a small constant factor greater than the original solver evaluation, regardless of the number of parameters, a vital feature for high-dimensional geoaoustic inversion problems. Furthermore, while symbolic differentiation also yields exact derivatives, it often suffers from “expression swell,” generating extremely complex symbolic formulas for the derivatives of intricate functions like physics simulators, which can be inefficient to evaluate and difficult to implement, especially for code involving control flow (loops, conditionals) (Corliss, 1988). AD avoids this by directly computing numerical derivative values, readily handling complex program structures, which are inherent in normal mode simulation. Thus, AD provides the necessary combination of accuracy, computational efficiency, practical applicability to complex code, and allows integration with gradient-based methods.

Recent work has begun to make classical acoustic propagation solvers differentiable. Chitre (2023) showed that a

Hamiltonian ray tracer can be embedded in an AD framework, enabling the gradient-based tuning of environmental parameters along eikonal paths and gradient-based source localization methods. Later, Lytaev (2024) derived a differentiable parabolic equation model, demonstrating the estimation of the real-time SSP using backpropagated gradients. Both studies focus on high-frequency forward models (rays or parabolic equation) and treat the acoustic field as a differentiable function of a one- or two-dimensional SSP. The present work extends the concept to normal mode theory: By differentiating the discretized eigenvalue problem, we obtain exact sensitivities of every modal quantity with respect to a full set of geoaoustic parameters and frequency. This complements ray- and parabolic equation-based approaches and opens gradient-based methods for low-frequency inversions, relevant to shallow water environments.

In this work, a differentiable implementation of an acoustic normal mode model is presented. The implementation maintains the computational accuracy and efficiency of traditional solvers while adding the capability to compute gradients of any output representation with respect to environmental parameters or frequency automatically. The utility of this approach is demonstrated by performing sensitivity analyses on classical benchmark problems. These analyses provide insight into parameter significance, validate our implementation, and offer a foundation for developing more efficient gradient-based inversion techniques. By enabling AD, this approach opens new possibilities for integrating physics-based acoustic propagation models with modern machine learning techniques, which could lead to more accurate and efficient solutions for inverse problems in underwater acoustics.

In what follows, Sec. II discusses AD, forward vs backward differentiation, derivatives of the acoustic normal modes, and calculating the group velocity. Section III presents the AD-based sensitivity analysis for three distinct ocean environments. The conclusions are discussed in Sec. IV.

II. DIFFERENTIABLE PROGRAM AND AUTOMATIC DIFFERENTIATION

To take advantage of AD for underwater acoustic modeling, a differentiable version of a finite-difference normal-mode solver was rewritten in the Julia programming language. This implementation is called Kraken.jl (Vardi, 2025) and is based on the well-established KRAKEN program originally developed by Michael Porter (Porter, 1992). Julia was chosen for its combination of high-performance numerical computing capabilities, native support for AD through multiple packages in its ecosystem, and the ease of use of its syntax and dynamic programming features. The Kraken.jl implementation is designed to be modular and extensible, allowing for easy integration with other Julia packages and libraries.

The Julia implementation maintains the main computational approach of the original KRAKEN while restructuring the code to be compatible with AD. The solver consists of

the following key stages (more details on each of these steps can be found in [Appendix A](#)):

1. Problem discretization: The acoustic wave equation is discretized using a FD scheme, resulting in a matrix eigenvalue problem for each horizontal wavenumber.
2. Wavenumber interval location: A bisection method is used to locate intervals in the horizontal wavenumber space where eigenvalues (modal wavenumbers) can be found. This step ensures that all propagating modes are identified and isolated.
3. Root finding: Within each interval identified in the previous step, a root-finding algorithm is applied to solve the precise modal wavenumbers $k_{r,m}$. Due to the modularity of the code, we utilize Brent's method provided by the *Roots.jl* package. Because the previous bisection step rigorously bounds each root, this bracketing method guarantees convergence and prevents the solver from encountering branch points.
4. Mode function calculation: For each modal wavenumber $k_{r,m}$, the corresponding modal depth function $\psi_m(z)$ is calculated using an inverse iteration technique, providing the vertical structure of the acoustic field for that mode.
5. Richardson extrapolation: To enhance accuracy, Richardson extrapolation is used by solving the problem multiple times with increasingly refined discrete meshes and combining the results to achieve higher-order accuracy for the modal wavenumber.

A. Forward vs reverse mode AD

The Julia language ecosystem offers several AD frameworks, including *ForwardDiff.jl* ([Revels et al., 2016](#)) for forward AD, *Zygote.jl* ([Innes, 2018](#)) for reverse AD, and more recently, *Enzyme.jl* for both forward and backward AD ([Moses and Churavy, 2020](#)). To understand how these calculate derivatives to machine precision, a program's execution can be viewed as a computational graph of intermediate variables, ν_i .

Forward-mode AD evaluates derivatives alongside the primal evaluation by carrying an extra dual component with every real number. A dual number can be represented as $x + \epsilon$, where ϵ is an infinitesimal such that $\epsilon^2 = 0$. By applying Taylor expansion, evaluating any differentiable function f at a dual number yields $f(x + \epsilon) = f(x) + f'(x)\epsilon$, automatically giving both the function value and its exact derivative. In the computational graph, forward mode propagates the derivative (the tangent), $\dot{v}_j = \partial v_j / \partial x$, from inputs to outputs using the chain rule as follows:

$$\dot{v}_j = \sum_i \frac{\partial v_j}{\partial v_i} \dot{v}_i. \quad (1)$$

where ν_i s and ν_j s are intermediate variables of the computational graph. This evaluates the derivative of all intermediate quantities with respect to a single input direction. Therefore, its computational cost scales linearly with

the number of independent variables, N . A key advantage is that it can compute derivatives of any specific section of the program without needing to evaluate unnecessary subsequent segments.

Reverse-mode AD (the adjoint) operates in two phases. First, it executes a standard forward pass, computing the primal variables and storing the execution trace. Then, it sweeps backward, accumulating sensitivities from the outputs. It computes the adjoint $\bar{v}_i = \partial y / \partial v_i$, which represents the sensitivity of the final scalar output y to the intermediate variable ν_i . The adjoints are accumulated using the reverse chain rule as follows:

$$\bar{v}_i = \sum_j \frac{\partial v_j}{\partial v_i} \bar{v}_j. \quad (2)$$

Because it accumulates backward from the result, its cost scales with the number of outputs, M , and is essentially constant for a scalar loss function. This makes it highly preferable for high-dimensional problems where $N \ll M$.

B. Derivatives of the normal mode solver

A key advantage that stems from the modular design of *Kraken.jl* (outlined in steps 1–5 above) combined with AD is the ability to compute derivatives of intermediate quantities within the solver pipeline. Since each stage (e.g., the root-finding process that yields the modal wavenumbers $k_{r,m}$) can be viewed as a differentiable function, AD frameworks such as *ForwardDiff.jl* can be used to compute its derivatives with respect to the input parameters without the need to execute subsequent steps, such as calculation of the mode function.

For example, one can efficiently calculate the sensitivity of the modal wavenumbers to environmental parameters (e.g., $\partial k_{r,m} / \partial c$, $\partial k_{r,m} / \partial \rho$, $\partial k_{r,m} / \partial h$) by applying AD only up to the root-finding stage. Here, $k_{r,m}$ are the modal wavenumbers, c is the compressional sound speed, ρ is the bulk density, and h is the layer thickness. This avoids the computational expense associated with the inverse iteration required to find the modal depth functions $\psi_m(z)$. This capability is useful for sensitivity analyses or optimization tasks that primarily depend on the wavenumbers themselves, rather than the full modal depth functions or derived quantities like the acoustic pressure field, which would require differentiating the complete solver chain. The calculation of group velocity using AD, discussed next, provides a concrete example on how to take advantage of this modular differentiability.

C. Calculating group velocity

In normal mode theory, the group velocity represents the speed at which the acoustic energy propagates in the waveguide. Traditionally, the group velocity for mode m has been calculated using perturbative methods as described in the seminal textbook *Computational Ocean Acoustics* ([Jensen et al., 2011](#), their Chap. 5). These methods require both the calculation of the modal wavenumbers $k_{r,m}$ and the

modal depth functions $\psi_m(z)$, and involve an expression of the following form:

$$v_{g,m} = \frac{\int_0^\infty \frac{1}{\rho(z)} \psi_m^2(z) dz}{\frac{\omega}{c^2(z)} \int_0^\infty \frac{1}{\rho(z)} \psi_m^2(z) dz}. \quad (3)$$

AD offers a more straightforward and efficient approach to calculate the group velocity. The group velocity is defined mathematically as follows:

$$v_{g,m} = \frac{d\omega}{dk_{r,m}}, \quad (4)$$

where $\omega = 2\pi f$ is the angular frequency. Using the differentiable implementation, the group velocity $\nu_{g,m}$ can be directly computed. Given that the program calculates $k_{r,m}$ as a function of frequency f , the relationship can be expressed as follows:

$$k_{r,m} = K_m(f), \quad (5)$$

where K_m is the function implemented by our differentiable normal mode solver that maps from frequency to wavenumber of mode m . The group velocity is then as follows:

$$v_{g,m} = \frac{2\pi}{dK_m/df}. \quad (6)$$

By applying AD to the function K_m , the derivative dK_m/df , accurate to machine precision, is obtained without any further calculation of the modal depth functions $\psi_m(z)$ and the numerical evaluation of the integral in Eq. (3). This approach requires only a single evaluation of the function with dual numbers, making it more accurate and computationally efficient than traditional methods.

The ability to accurately and efficiently compute group velocities using AD is relevant for several widely used geoaoustic inversion techniques. Methods based on analyzing modal dispersion, such as time warping, are based directly on the frequency-dependent nature of group velocities $\nu_{g,m}(f)$ to interpret the arrival structure of broadband acoustic signals recorded on a single hydrophone (Bonnell and Chapman, 2011; Bonnell *et al.*, 2020a; Bonnell *et al.*, 2020b; Gervaise *et al.*, 2012; Petrov, 2014). These techniques extract information about the environment by matching observed dispersion curves, which are linked to group speeds, with replicas generated by forward models. Therefore, having a precise and computationally inexpensive method like AD to calculate $\nu_{g,m}$ (and potentially its sensitivities to environmental parameters) is advantageous for the implementation and performance of such inversion schemes.

III. SENSITIVITY ANALYSIS

Typically, a sensitivity analysis is performed to understand the effects of changes in geoaoustic parameters on

key model output [e.g., modal wavenumbers $k_{r,m}$ and modal depth functions $\psi_m(z)$ in our case] that directly influence the predicted acoustic field. This analysis helps identify which parameters have the most significant impact on acoustic predictions, guide experimental design, prioritize inversion parameters, and evaluate model uncertainty (Dosso *et al.*, 2007a; Dosso *et al.*, 2007b). By understanding these sensitivities, researchers can gain insight into the robustness and reliability of inversion results and ensure that the inversion process focuses on the most influential parameters.

To showcase the potential of AD for gradient calculation, a sensitivity analysis is performed on three representative benchmarks.

1. Pekeris waveguide (Pekeris, 1948): This canonical shallow-water environment provides a crucial validation case. Because it possesses an analytical solution, it allows for direct comparison and verification of the sensitivities computed via AD against known theoretical results.
2. New England Mudpatch environment: Representing a complex, realistic shallow-water scenario derived from recent experiments (Wilson *et al.*, 2020, 2022), this benchmark tests the AD approach on a multilayered geoaoustic profile. It provides information on the application of sensitivity analysis to practical inversion problems in environments with significant seabed interaction.
3. Munk profile (Munk and Wunsch, 1979): This classic deep-water SSP introduces a different acoustic regimen characterized by a prominent acoustic channel and a large number of propagating modes. Analyzing the Munk profile demonstrates AD's capability to handle sensitivity calculations in acoustically complex, deep-ocean scenarios.

These 3 underwater environments are described using a set of geoaoustic parameters, which are presented in Fig. 1. The main parameters describing each model are the sound speeds c , the bulk density ρ , and the layer thickness h . All parameters describing the water columns are subscripted with the letter w , and all parameters describing the bottom half-space are subscripted with the letter b .

A. Pekeris waveguide

The Pekeris waveguide is frequently used in underwater acoustics modeling due to its simplicity and ability to capture essential wave propagation characteristics in a stratified medium. Moreover, the Pekeris waveguide has an analytical solution, which can be used to validate the output from the numerical solvers. The waveguide consists of a water column with constant sound speed that overlays a bottom half-space with different set of acoustic properties. The geoaoustic parameter values used from now on for the Pekeris waveguide are given in Fig. 1(a). The water column layer has a constant sound speed value of 1500 m/s and a bulk density of 1000 kg/m³, and the bottom half-space has a sound speed value of 1600 m/s and a bulk density of 1500 kg/m³.

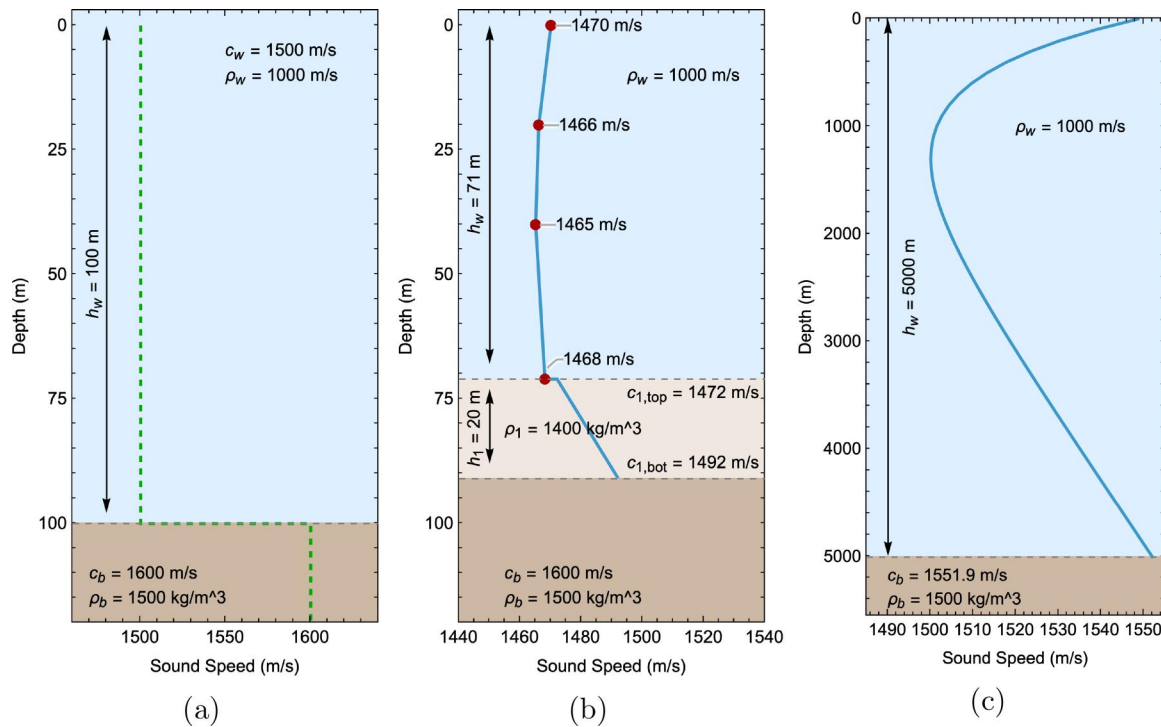


FIG. 1. Geoacoustic models used for testing and validation of the differentiable normal-mode solver: (a) Pekeris waveguide; (b) New England Mudpatch; and (c) Munk profile.

The three main expressions that are compared are the resulting modal depth functions $\psi_m(z)$, their associated wavenumbers $k_{r,m}$ and the group velocity $\nu_{g,m}$ of each mode. The geoacoustic model shown in Fig. 1(a) was used as input to both Kraken.jl and KRAKEN at a frequency of 100 Hz. The results for the first five modal wavenumbers $k_{r,m}$ are shown in Table I and the modal depth functions are shown in Fig. 2. Both results are identical using all three methods, confirming that the Julia implementation accurately reproduces the correct solutions.

The group velocity results, shown in Fig. 3, further validate the Julia implementation against both the original KRAKEN and the analytical solution for the Pekeris waveguide. Excellent agreement is observed between all three methods, including with the Airy phase minima, which confirms the accuracy of the AD-based calculation [Eq. (4)] where the underlying modal solution is well resolved. However, a limitation of the numerical solver becomes apparent at low frequencies as the modes approach their

cutoff frequency. The numerical solutions (for Kraken.jl and KRAKEN) appear to terminate prematurely compared to the analytical prediction, which shows $\nu_{g,m}$ correctly approaching the bottom sound speed c_b for lower frequencies. This suggests numerical challenges in accurately resolving the modal properties and their derivatives using a discrete grid very close to the mode's cutoff point. Although the AD approach correctly differentiates the available numerical results, it inherits this underlying limitation near the cutoff frequency, but this limitation is different than that of the FD approach. Derivatives calculated using AD are derivatives of the algorithm itself, thus, will inherit any numerical error that the algorithm produces on the original solution (such as spatial discretization errors near the cutoff frequency or deviation from the true solution due to not meeting convergence criteria). In the FD case, numerical errors on the derivatives are exacerbated due to truncation and rounding errors, both of which are alleviated by using AD.

The sensitivity analysis of the modal depth functions with respect to environmental parameters provides valuable insights into which parameters most strongly influence the acoustic propagation. Figure 4 compares the derivatives of the modal depth functions calculated using three different approaches: FD approximation, AD, and analytic differentiation. The results show excellent agreement between the derivatives computed using the analytical solution and AD. On the other hand, the FD method was subject to errors, especially for the 5th mode, due to incorrect step size choice, which was chosen to be a universal relative step size

TABLE I. Comparison of wavenumbers calculated at 100 Hz with the Pekeris waveguide using a FORTRAN and Julia versions of KRAKEN along with the analytic solution.

Mode no.	k_r (Julia)	k_r (FORTRAN)	k_r (Analytic)
1	0.417 908	0.417 908	0.417 908
2	0.414 963	0.414 963	0.414 963
3	0.409 968	0.409 968	0.409 968
4	0.402 850	0.402 850	0.402 850
5	0.393 808	0.393 808	0.393 808

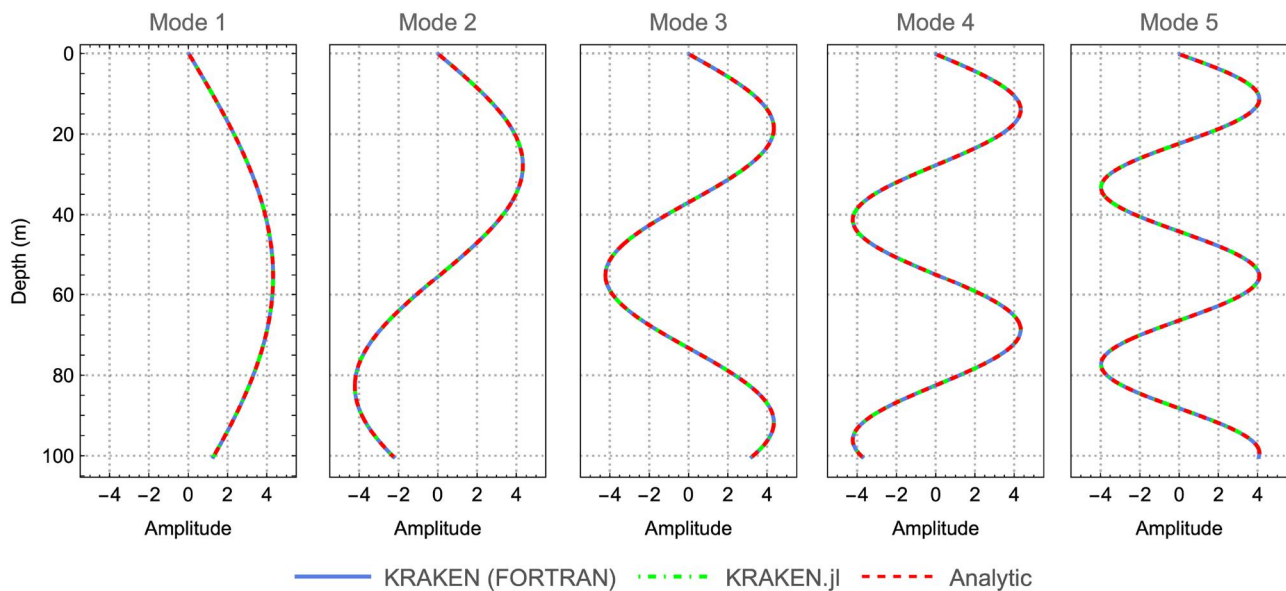


FIG. 2. Modal depth functions calculated using a FORTRAN and Julia versions of KRAKEN along with the analytic solution using the Pekeris waveguide 100 Hz.

of 1%. This highlights the fragility of depending on FD for accurate gradient calculations without conducting a convergence analysis prior to using this method for each environmental parameter separately.

The derivatives of the modes with respect to the environmental parameters, computed using AD, are consolidated in the right column of Fig. 4 for easier analysis. Several features are apparent. The magnitude of sensitivity generally increases with the mode number, with mode 5 exhibiting the largest shape changes for perturbations in water sound speed c_w , bottom sound speed c_b , and water depth h_w . This increased sensitivity for higher-order modes, which are often closer to cutoff (Jensen *et al.*, 2011), is a known phenomenon and can make them particularly informative, yet challenging, for inversion (Zanolin *et al.*, 2004).

Furthermore, the derivatives with respect to the water column sound speed c_w and the bottom sound speed c_b show

somewhat mirrored patterns, especially near the seabed interface. This suggests that changes in these two parameters can have opposing effects on the mode shapes, potentially leading to parameter correlations or ambiguities in inversion if not carefully handled. For example, an increase in c_w could push the modal energy downward, while an increase in c_b could confine it more strongly in the water column, leading to complex interactions reflected in the derivative shapes.

A qualitative comparison between AD and FD is also provided in Table II for the Pekeris waveguide. Table II shows that computing modal function sensitivities with forward-mode AD using Kraken.jl costs only 482 μ s, roughly doubling a forward-only run yet still delivering all five parameter derivatives in a single sweep. A central FD scheme with the legacy FORTRAN KRAKEN, by contrast, needs ten perturbed evaluations, inflating the runtime to 1320 μ s, which is about 11 times its own forward pass and $\sim 2.7\times$ slower than AD. Although both approaches scale linearly with the number of parameters, forward-mode AD carries a constant factor of $1 \times N$ versus $2 \times N$ for FDs and eliminates the step-size tuning and convergence checks required by the latter, making it the more efficient and robust choice for larger sensitivity analyses.

B. New England Mudpatch

The following geoacoustic model represents the New England Mudpatch sediment structure, which is a more realistic case for underwater environments. Here, the SSP of the water column resembles the profile measured during the SBCEX22 experiment in the New England Mudpatch (Vardi *et al.*, 2024). The values of the geoacoustic parameters used are shown in Fig. 1(b). The water column sound speed is defined at four depth points of 0, 20, 40, and 71 m and their corresponding sound speed values at those depths

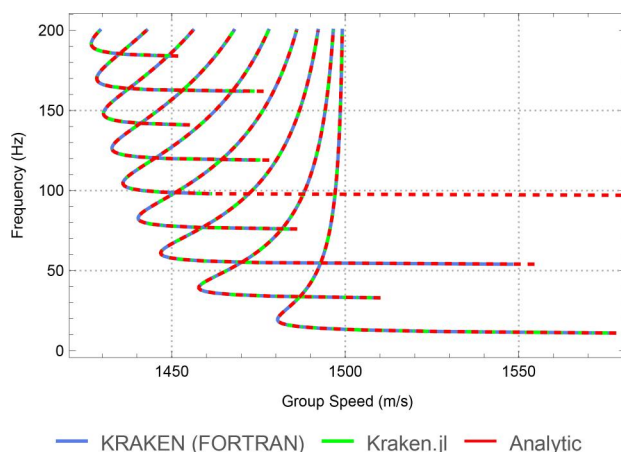


FIG. 3. Calculated group speeds for the Pekeris waveguide using both versions of Kraken and an analytical solution.

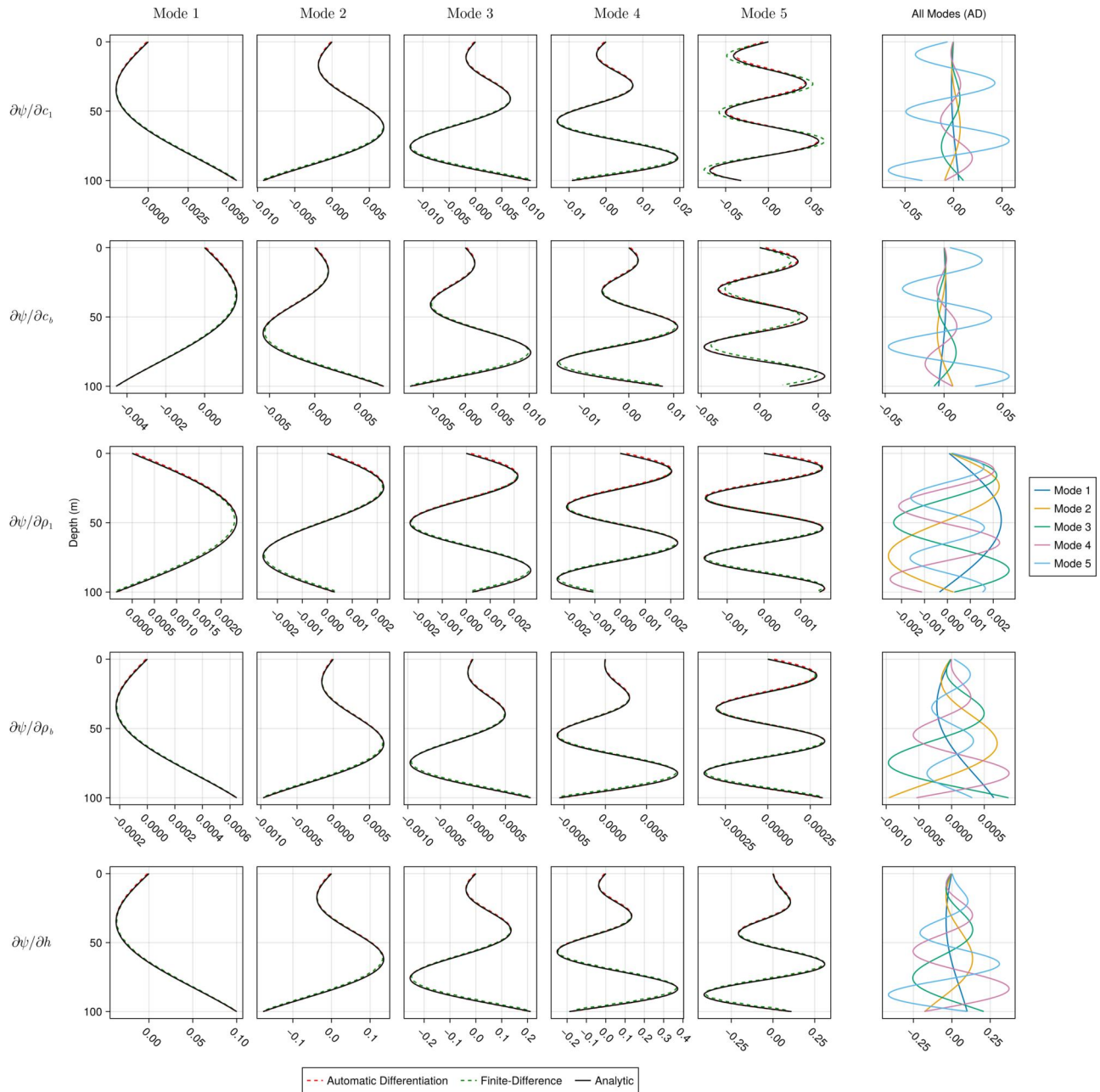


FIG. 4. Derivatives of the modal depth functions using three different methods for the Pekeris waveguide: FD, AD, and analytic differentiation. Combined results of all modal derivatives with respect to each parameter are given at the rightmost column using the AD approach.

are c_{w0} , c_{w1} , c_{w2} , and c_{w3} . Using this environment, there is no analytical solution. Thus, a comparison can only be made with an existing well-tested solver, which in this case is the original KRAKEN.

At a frequency of 100 Hz, the wavenumbers and the modal depth functions are calculated using the Julia and FORTRAN implementations of the program, using the mud-patch environment as input. The wavenumber solutions

TABLE II. Computational cost of forward-mode AD versus central FDs for modal-function derivatives.

Aspect	Forward-mode AD (Kraken.jl)	Central FD (FORTRAN KRAKEN)
Extra model evaluations	1 pass (no perturbations)	$2N = 10$ perturbed runs
Time to obtain all derivatives	$482 \mu\text{s}$	$1320 \mu\text{s}$
Cost relative to forward run	$\approx 2\times$	$\approx 11\times$
Speed-up (AD vs. FD)	—	AD is $\sim 2.7\times$ faster
Scaling with no. of parameters	$O(N)$ (one sweep)	$O(N)$ (two sweeps)
Practical overhead	None (no step size)	Step-size tuning, convergence checks

TABLE III. Comparison of wavenumbers calculated at 100 Hz using the New England Mudpatch environment as input to a FORTRAN and Julia versions of KRAKEN.

Mode no.	k_r (Julia)	k_r (FORTRAN)
1	0.427 239	0.427 332
2	0.422 798	0.422 872
3	0.415 695	0.415 755
4	0.407 484	0.407 547
5	0.397 212	0.397 289

using each program can be found in the Appendix under Table III, while the corresponding modes are illustrated in Appendix A.

In this environment, we have increased the number of geoaoustic parameters describing it to 12 parameters overall, compared to 5 for the Pekeris waveguide. Moreover, compared to the Pekeris waveguide, it has an additional sediment layer, which adds complexity to the environment. Using AD, the gradients of the modal depth functions are calculated with respect to the environmental parameters and are shown in Fig. 5. Several interesting observations can be seen here.

First, the derivatives with respect to the water column sound speed parameters c_{w0} , c_{w1} , c_{w2} , and c_{w3} [which are defined in Fig. 1(b)] are significant across all modes, sometimes exceeding the sensitivity to sediment sound speeds (c_1). This highlights why accurate measurements of the SSP throughout the water column are essential for reliable geoaoustic inversion, as also indicated by studies involving joint inversion of the water column and the properties of the seabed (Dosso and Bonnel, 2023).

Second, the parameters of the mud layer c_1 , ρ_1 , and h_1 show strong sensitivities, near the water-sediment interface and within the mud layer itself. The mud sound speed c_1 has

large derivatives, confirming its strong influence on the acoustic field in this shallow water environment, consistent with previous New England Mudpatch inversion studies that identified it as a key parameter (Bonnel *et al.*, 2024). In contrast, the bottom half-space parameters c_b and ρ_b show smaller derivative magnitudes compared to the mud layer parameters, especially for the lower order modes, such as modes 1 and 2. This reduced sensitivity suggests that, at this frequency of 100 Hz, the deeper bottom properties have less influence, especially for modes that are mainly confined within the water and mud layer. However, consistent with physical intuition, modes 3 and higher, which interact more strongly with the bottom due to steeper propagation angles, show greater sensitivity to these bottom parameters compared to the lower-order modes.

C. Munk profile

The Munk profile is a deep-water SSP that has been well studied in the past. The geoaoustic model that describes the Munk profile is depicted in Fig. 1(c). Unlike shallow-water cases, the Munk environment is characterized by a deep sound channel that results in significantly more propagating modes. The modes and wavenumbers for this model were calculated at 100 Hz using both Julia and FORTRAN implementations. Some representative modes chosen from the 204 propagating modes are shown in Fig. 8, demonstrating strong agreement between the implementations.

What distinguishes the Munk profile analysis is the complex interplay between the detailed sound speed structure and the modal propagation characteristics in deep water. To investigate this, we examine the sensitivity of the modal wavenumbers to perturbations throughout the depth-dependent SSP $\partial k_{r,m}/\partial c_j$ where c_j is the water sound speed at the depth discretization point j . This particular quantity is informative because it reveals which depth regions most

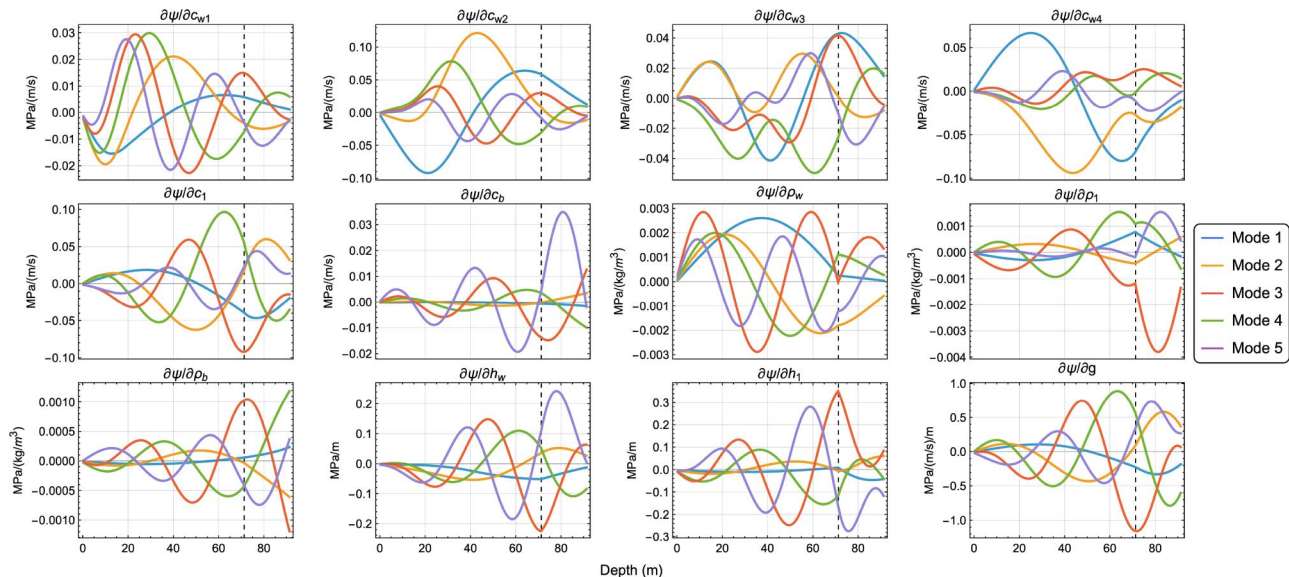


FIG. 5. Derivatives of the mode functions with respect to each of the environmental parameters defining the New England Mudpatch model as seen in Fig. 1(b).

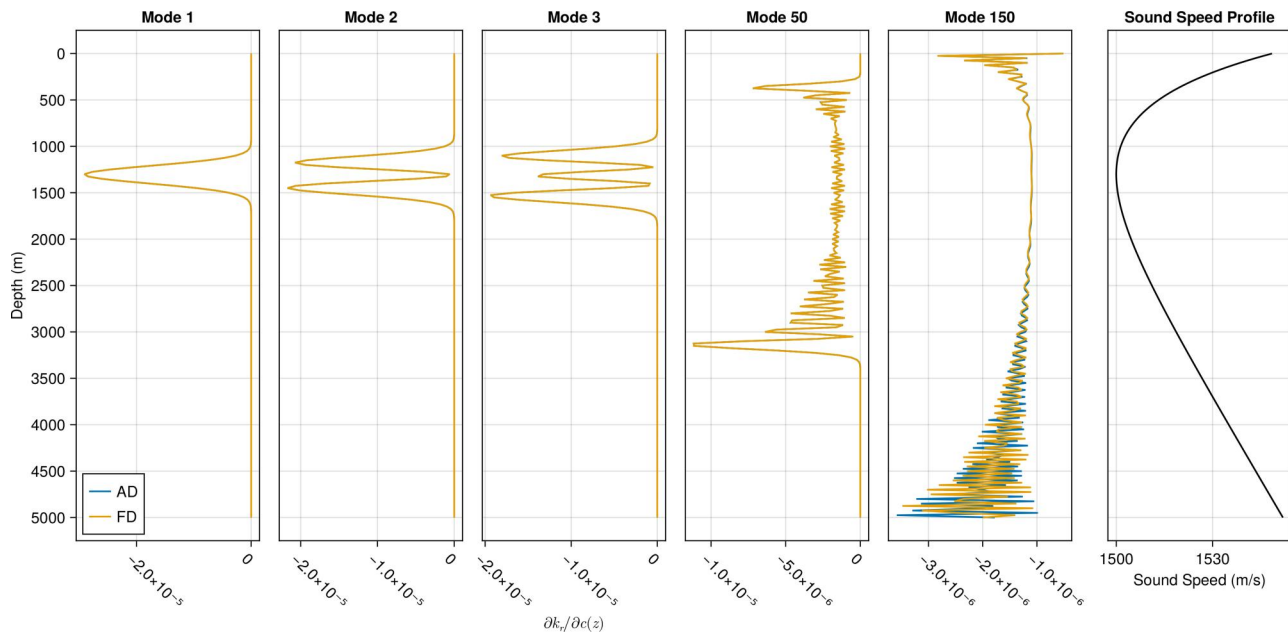


FIG. 6. The derivative of the wavenumbers for the first three modes, mode 50, and mode 150 as a function of the depth-dependent Munk SSP, calculated using AD and FD approaches.

strongly influence the phase velocity and horizontal propagation characteristics of each mode. The derivative of all 204 propagating wavenumbers is shown in Appendix B (see Fig. 9), while here we will focus on five different modes for clarity, the modes being modes 1, 2, 3, 50, and 150 shown in Fig. 6.

The oscillatory patterns visible in these derivatives reveal which depth regions most strongly influence acoustic propagation, directly related to the oscillatory nature of the modal functions. These oscillations arise from constructive and destructive interference patterns that form within the waveguide. The negative values arise from the inverse relationship between phase speed and wavenumber, where the phase speed is directly affected by the “fastness” of the waveguide, which is defined from the sound speed of the medium. Each mode exhibits maximum sensitivity between specific depths corresponding to where that mode concentrates its energy.

Moreover, we can see good agreement between the derivatives obtained using the AD and FD methods, with deviations seen mostly in mode 150. In mode 150, the oscillations are more pronounced, especially near the top and bottom of the water column. To obtain better agreement, a more thorough convergence analysis should be performed prior to using FD to minimize the error between the results obtained using AD (considered as the accurate and true derivative). This is another example in which the FD approach for calculating sensitivities can introduce numerical errors if the perturbation step size is not properly chosen.

IV. CONCLUSION

This study details the development and validation of Kraken.jl, a differentiable implementation of the KRAKEN normal mode model built using the Julia language and AD.

By successfully reproducing the results of the original FORTRAN code and analytical benchmarks (Pekeris waveguide), the core accuracy of the forward solver was confirmed. Furthermore, the application of AD enabled efficient and accurate computation of sensitivities—derivatives of modal properties (wavenumbers, mode shapes) with respect to environmental parameters and frequency—as demonstrated through analyses of the Pekeris, New England Mudpatch, and Munk profile environments. These sensitivity results not only validate the implementation but also provide physical insights, such as identifying key parameters influencing propagation in different acoustic regimes (e.g., mud layer properties in New England Mudpatch environment, sound channel structure in the Munk profile).

A key advantage demonstrated was the computational efficiency gained through AD, particularly when combined with the solver’s modularity. Calculating sensitivities via AD proved significantly faster than traditional FD approaches, especially for problems involving multiple parameters. While using the forward-mode AD still requires N calculations for a gradient in N parameters, it does not require performing prior convergence analysis for step-size determination, as needed when calculating gradients using a FD approach. Moreover, any gradient calculated using AD is guaranteed to be accurate to machine precision. This efficiency opens practical pathways for gradient-based optimization techniques in geoacoustic inversion, potentially offering faster convergence and more robust solutions compared to conventional sampling or linearized methods. The availability of accurate gradients also holds promise for more sophisticated uncertainty quantification within Bayesian frameworks.

Reverse mode (implemented with Enzyme.jl) has been added up to the wavenumber root-finding stage so that the

gradient $\nabla_{\theta} k_{r,m}$ with respect to environmental parameters and frequency θ is available in a single reverse pass. Extending the adjoint beyond that point to the full pressure-field calculation was left for future work due to its implementation complexity, while forward mode is already fast enough for sensitivity analysis purposes. Both directions will result in the same accurate gradient calculation, and the direction chosen is a matter of performance consideration depending on the application. In this work, forward mode (via ForwardDiff.jl) is used for most examples because the parameter sets of interest (frequency and geoacoustic properties) are few while output dimensions are large.

The approach taken here, making a well-established physics-based solver directly differentiable, contributes to the broader field of scientific machine learning and differentiable programming (Thuerey *et al.*, 2022). It represents one strategy for merging physical knowledge with data-driven techniques. Another prominent strategy is the use of physics-informed neural networks (Cuomo *et al.*, 2022), where physical laws (often partial differential equations or ordinary differential equations) are incorporated into the loss function of the neural network during training (Yoon *et al.*, 2024a; Yoon *et al.*, 2024b). While physics-informed neural networks learn solutions within a neural network framework, the differentiable solver approach maintains the structure of the original physics model, allowing for direct computation of sensitivities from that model. Both paradigms leverage AD to bridge physics and machine learning, offering complementary tools to tackle complex scientific problems. In conclusion, the development of Kraken.jl provides a validated, efficient, and flexible tool for normal mode modeling and sensitivity analysis. It serves as a foundation for future work in geoacoustic inversion, uncertainty quantification, and the development of hybrid physics-ML models for underwater acoustics.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The Julia version of KRAKEN can be found at <https://github.com/vardister/Kraken.jl>.

APPENDIX A: WRITING A NORMAL-MODE CODE

A normal mode code, KRAKEN.jl, aims to rewrite the well-known FORTRAN program of the same name from the Acoustics Toolbox. Michael Porter originally wrote the

FORTRAN program using normal-mode theory to solve underwater acoustic fields. By rewriting it in Julia, we gain access to modern tooling and interactive runtime without performance loss. AD is a key benefit that enables program differentiation. The new program is inspired by Porter's original FORTRAN version and Hunter Atkins's Python implementation *pykrak*. The solution process follows the Jensen *et al.* (2011) book. The Julia implementation follows these steps:

1. Define the FD matrix for the acoustic problem.
2. Use the bisection method to find isolating intervals where wavenumbers $k_{r,m}$ can be found and then use a root finding algorithm to solve for the wavenumbers $k_{r,m}$ in each interval.
3. Perform an inverse iteration using the wavenumbers found $k_{r,m}$ to solve the modal depth functions $\psi_m(z)$.
4. Use Richardson's extrapolation using finer discret meshes to improve the wavenumber approximation up to a degree of accuracy defined by the user.

A. Defining the FD matrix

We seek to solve the separated Helmholtz equation in cylindrical coordinates. More specifically, we want to solve for the modal depth functions $\Psi_m(z)$ and their associated wavenumbers $k_{r,m}$ using the following equation:

$$\rho(z) \frac{d}{dz} \left(\frac{1}{\rho(z)} \frac{d\Psi_m(z)}{dz} \right) + \left(\frac{\omega^2}{c^2(z)} - k_{r,m}^2 \right) \Psi_m(z) = 0. \quad (\text{A1})$$

Once found, the modal field solution for the pressure p_f is as follows:

$$p_f(r, z) = \frac{i}{\rho(z_s) \sqrt{8\pi r}} e^{-i\pi/4} \sum_{m=1}^{\infty} \Psi_m(z_s) \Psi_m(z) \frac{e^{ik_{r,m}r}}{\sqrt{k_{r,m}}}, \quad (\text{A2})$$

where r is the source-receiver range, z_s the source depth and z the receiver depth. To solve Eq. (A1), a FD scheme is used in order to transform the problem into an algebraic eigenvalue problem of the following form:

$$(\mathbf{A} - \lambda_m(k_{r,m}) \mathbf{I}) \psi_m = 0, \quad (\text{A3})$$

where matrix $\mathbf{A}$ is a tridiagonal matrix of components stemming from a finite differencing Eq. (A1) and $\bar{\lambda}$ is a diagonal matrix of components λ_k which are a function k_r and its coefficient such that

$$\lambda_k = \frac{k_r^2 h_k^2}{h_k \rho_k}, \quad (\text{A4})$$

and index k refers to the layer number of the geoacoustic model. Solving this problem will result in finding the eigenvalues $\lambda_{k,m}$, and in turn the modal wavenumbers $k_{r,m}$.

Following the FD derivation in (Jensen *et al.*, 2011, their Chap. 5), matrix $\mathbf{A}$ gets to be defined as follows:

$$\mathbf{A} = \begin{pmatrix} a_0(\lambda) & e_1 & & & & \\ e_1 & a_1 & e_2 & & & \\ & e_2 & a_2 & e_3 & & \\ & & \ddots & \ddots & \ddots & \\ & & & e_{N-2} & a_{N-2} & e_{N-1} \\ & & & & e_{N-1} & a_{N-1} & e_N \\ & & & & & e_N & a_N(\lambda) \end{pmatrix}, \quad (\text{A5})$$

where each diagonal value a_j is defined by the following term for every depth point that is not on an interface as follows:

$$a_j = \frac{-2 + h_k^2 \omega^2 / c^2(z_j)}{h_k \rho_k}, \quad (\text{A6})$$

and off-diagonal values are as follows:

$$e_j = \frac{1}{h_k \rho_k}, \quad (\text{A7})$$

where index j is the discrete depth value defined by $z_j = j \Delta z_k$.

Here, we assume that the top boundary condition is vacuum, and thus the top derivative is infinity, and we remove the first row and column of matrix $\mathbf{A}$. The bottom half-space boundary condition is written as follows:

$$a_N = \frac{-2 + h_k^2 \omega^2 / c^2(z_N)}{2 h_k \rho_k} - \frac{\sqrt{k_r^2 - \omega / c_b}}{\rho_b}, \quad (\text{A8})$$

where c_b and ρ_b are the sound speed and density of the bottom half-space, respectively. Notice that the final entry a_N is a function of the wavenumber k_r which has yet to be found. Thus, the value is updated every time matrix $\mathbf{A}$ is being used to solve for the wavenumber itself.

Typically, geoacoustic models have multiple layers defined by changes in bulk density values ρ . To incorporate these interfaces into the FD matrix, the continuity of pressure and normal velocity must be respected. This leads to the following expression for the diagonal value a_i located at the interface between layer k and $k+1$ as follows:

$$a_j = \frac{-1 + \omega^2 h_k^2 / c_j^2}{2 h_k \rho_k} + \frac{-1 + \omega^2 h_{k+1}^2 / c_{j+1}^2}{2 h_{k+1} \rho_{k+1}}.$$

B. Solving for the wavenumbers

Once the FD matrix $\mathbf{A}$ has been created (excluding the last diagonal entry), Eq. (A3) can be solved by finding for which values k_r the determinant is zero such that

$$|\mathbf{A}(k_r) - \lambda^2(k_r) \mathbf{I}| = 0. \quad (\text{A9})$$

The solution for all wavenumbers $k_{r,m}$ involves a two-step process: isolating intervals where the wavenumbers can

be found and then using a root finding algorithm to solve Eq. (A9) for each wavenumber $k_{r,m}$ in their corresponding interval.

First, to isolate the intervals, a bisection method is used together with the Sturm's sequence p_n , which is defined as follows:

$$p_0 = 0, \quad (\text{A10})$$

$$p_1 = 1, \quad (\text{A11})$$

$$p_n = (\lambda_{n-1} - a_{n-1})p_{n-1} - e_{n-1}^2 p_{n-2}. \quad (\text{A12})$$

The sequence has the property of having M zero-crossings corresponding to the number of wavenumbers above a trial wavenumber used for calculating the sequence. Using this property, the bisection method can be used to isolate the intervals (for more information, see Jensen *et al.*, 2011, their Chap. 5.7).

The sequence is also used to calculate the determinant in Eq. (A9) knowing that the value p_n corresponds to the determinant value of the $n \times n$ matrix. Therefore, to isolate the wavenumbers $k_{r,m}$ in each interval, we solve for them using the following equation:

$$p_{N+1}(k_{r,m}) = 0, \quad k_{r,m} \in (k_{r,m}^-, k_{r,m}^+), \quad (\text{A13})$$

where $(k_{r,m}^-, k_{r,m}^+)$ is the interval for each wavenumber $k_{r,m}$ in the problem. Here, we use Brent's method via the Roots.jl package. Because the Sturm sequence strictly isolates exactly one root per interval, this bracketing approach is unconditionally stable and avoids the branch cut issues that can plague open-domain root-finding algorithms.

C. Inverse iteration

After finding all possible real wavenumbers, an inverse iteration process is performed to find the corresponding modal depth functions ψ_m (eigenfunction) for each wavenumber. This process is also highly efficient. It starts by assuming that the eigenvalue found $\tilde{\lambda}_m$ is an estimate of the true eigenvalue such that $\tilde{\lambda}_m = \lambda_m - \epsilon$ and ϵ is small. The iteration process involves solving for $\mathbf{w}_k$ as $k \rightarrow \infty$ as follows:

$$(\mathbf{A}(\tilde{\lambda}_m) - \tilde{\lambda}_m \mathbf{I}) \mathbf{w}_k = \mathbf{w}_{k-1}, \quad k = 1, 2, \dots, \infty. \quad (\text{A14})$$

This is done by acknowledging that the eigenvectors of the inverse matrix $(\mathbf{A}(\tilde{\lambda}_m) - \tilde{\lambda}_m \mathbf{I})^{-1}$ have the same eigenvectors as for matrix $\mathbf{A}$, and thus the solution for the modal depth function is as follows:

$$\mathbf{w}_k \rightarrow \psi_m, \quad k \rightarrow \infty. \quad (\text{A15})$$

The wavenumbers themselves can also improve through this process due to the limiting property as follows:

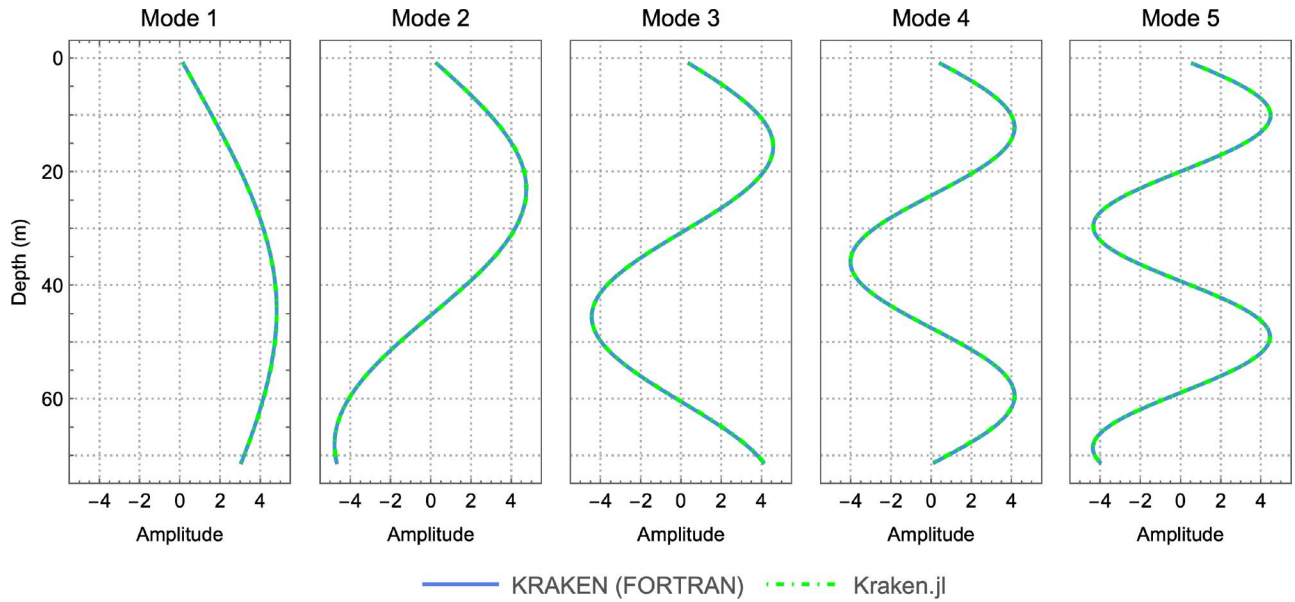


FIG. 7. Modal depth functions calculated at 100 Hz using a FORTRAN and Julia versions of KRAKEN using the New England Mudpatch environment as input.

$$\lambda_m \leftarrow \tilde{\lambda}_m + \frac{(\mathbf{w}_r)_{k-1}}{(\mathbf{w}_r)_k}, \quad k \rightarrow \infty. \quad (\text{A16})$$

Updating the eigenvalue estimate at each step yields cubic convergence, minimizing the required number of iterations (Trefethen and Bau, 2022).

Another point to be careful with is that, if the initial estimate for λ_m is close to the real eigenvalue, the inverse iteration process might blow up numerically since

$$(\mathbf{A}(\lambda_m) - \tilde{\lambda}_m \mathbf{I}) \mathbf{w}_k \rightarrow 0, \quad (\text{A17})$$

as $\tilde{\lambda}_m \rightarrow \lambda_m$ and thus we require that $|\mathbf{w}_k| = 1$ after each iteration step.

D. Richardson's extrapolation

The last step for improving the wavenumber estimation is to use Richardson's extrapolation. Knowing that the eigenvalue estimation varies as a function of the mesh width Δz , the extrapolation is expressed as follows:

$$k_r^2(\Delta z) = k_0^2 + b_2 \Delta z^2 + b_4 \Delta z^4 + \dots, \quad (\text{A18})$$

where k_0^2 denotes the exact eigenvalue of the continuous problem. The pseudo-eigenvalue problem is solved using a series of meshes and then a polynomial is fitted through Δz^2 . Then, Eq. (A18) is evaluated at $\Delta z = 0$ to obtain the extrapolation of the wavenumber value $k_r^2(0) = k_0^2$.

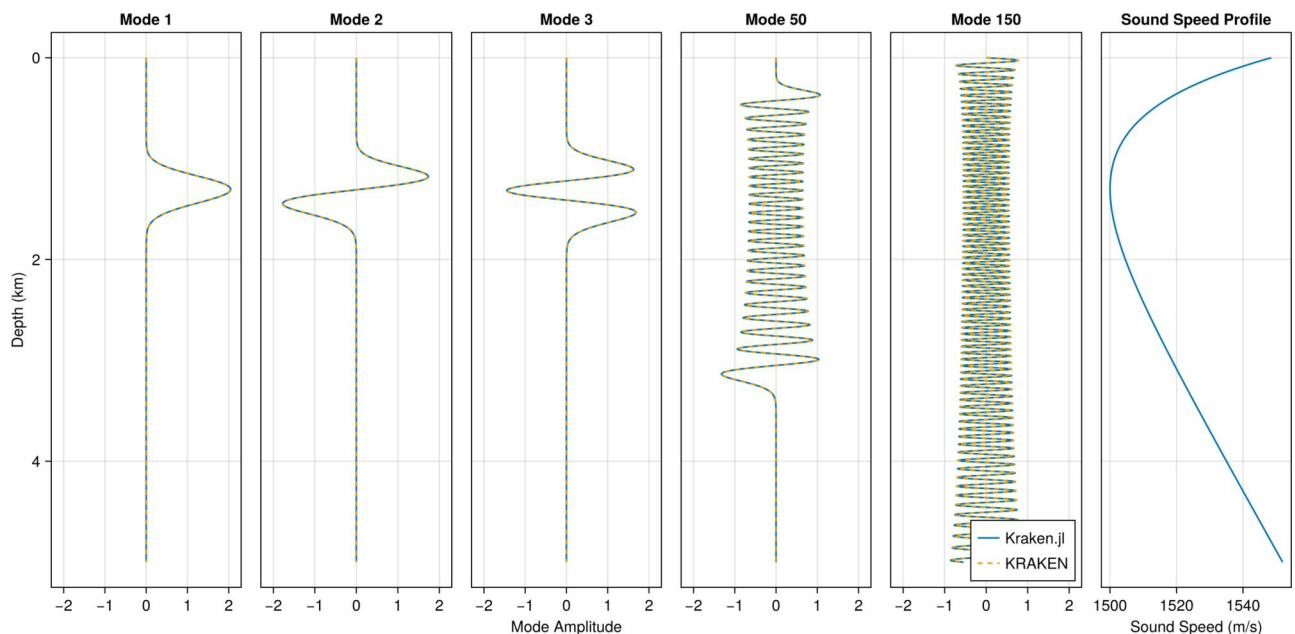


FIG. 8. Different modal depth functions for the Munk profile environment depicted in Fig. 1(c) using both Kraken.jl (Julia) and KRAKEN (FORTRAN).

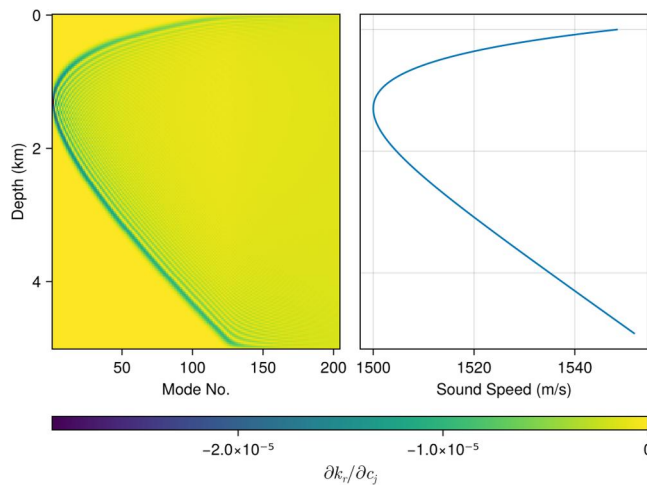


FIG. 9. The derivative of the wavenumber for each mode as a function of the depth-dependent Munk SSP.

APPENDIX B: MODEL OUTPUT COMPARISON BETWEEN JULIA AND FORTRAN IMPLEMENTATIONS

The following are an assortment of calculations done using both Julia and FORTRAN versions of Kraken for the New England Mudpatch and the Munk profile underwater environments used in the article.

1. New England Mudpatch (see Fig. 7).
2. Munk profile (see Figs. 8 and 9).

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